

LABORATORY OBSERVATION

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 2

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1. TITLE

Design of a Webpage using CSS3 Styling and HTML5 Canvas Graphics

2. AIM

To develop an interactive webpage utilizing CSS3 styling properties (flexbox container, cards, hover effects, custom typography, linear gradient backgrounds) and HTML5 Canvas API to render 2D graphics (shapes, lines, and text).

3. OBJECTIVE

The objectives of this experiment are:

- To apply CSS3 styling including gradients, borders, shadows, and hover pseudo-classes.
- To implement responsive layout design using CSS Flexbox.
- To render 2D graphic shapes (rectangles, circles, lines) using HTML5 Canvas JavaScript API.
- To customize typography and interactive card components.

4. THEORY

CSS3 (Cascading Style Sheets Level 3) extends basic styling with advanced capabilities such as flexbox layouts, box shadows, gradients, and pseudo-class state selectors (:hover). HTML5 <canvas> provides a drawable region in HTML accessed via JavaScript's CanvasRenderingContext2D interface to render dynamic shapes, paths, text, and graphics directly in the browser window.

5. ALGORITHM / PROCEDURE

1. Create index.html and set up standard document structure with embedded <style> block.
2. Define global CSS rules for reset, typography, and page background gradient.
3. Style navigation bar links with borders, padding, and hover state transitions.
4. Add HTML5 <canvas> element and script context initialization.
5. Draw 2D canvas shapes (filled rectangle, filled circle, stroke line) and custom text.
6. Design container with Flexbox layout (display: flex) for responsive alignment.
7. Create styled card components with box shadows, rounded borders, and hover effects.
8. Embed action buttons within card layouts with customized background and hover styles.
9. Test and verify webpage rendering and canvas graphics in a browser.

6. CODE EXPLANATION

The HTML document begins with document type declaration and contains an embedded <style> block configuring CSS3 rules. Linear gradients (background: linear-gradient) and font styling are applied to the body and header. The <nav> bar utilizes anchor tags with padding, borders, and :hover background transitions. The <canvas> element is scripted via JavaScript using getContext("2d") to draw a filled blue rectangle, a red circle (arc), a green line path (lineTo), and text ("Nandu"). A flex container (.container) holds three feature cards (.card) styled with box-shadows, borders, and hover elevation effects.

7. PROGRAM

index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>My Styled Webpage</title>
  <style>
    * {
      margin: 0;
      padding: 0;
    }
    body {
      font-family: Arial, Helvetica, sans-serif;
      background: linear-gradient(white, #e8e8e8);
      padding: 15px;
    }
    header {
      text-align: center;
      padding: 10px;
      font-family: Georgia, serif;
    }
    header h1 {
      font-size: 32px;
    }
    header p {
      font-size: 16px;
      font-family: Arial, sans-serif;
    }
    nav {
      text-align: center;
      padding: 10px;
    }
    nav a {
      color: black;
      text-decoration: none;
      padding: 6px 16px;
      margin: 0 5px;
      border: 1px solid #ccc;
      border-radius: 5px;
    }
    nav a:hover {
      background: #ddd;
      color: #333;
      border-color: #999;
    }
    canvas {
      border: 2px solid #333;
      border-radius: 8px;
    }
```

```

        display: block;
        margin: 10px auto;
    }
    h2 {
        text-align: center;
        font-family: Georgia, serif;
    }
    .container {
        display: flex;
        padding: 10px 0;
    }
    .card {
        border: 2px solid #666;
        border-radius: 10px;
        padding: 15px;
        width: 180px;
        margin: 5px;
        box-shadow: 3px 3px 8px #aaa;
    }
    .card h3 {
        font-size: 20px;
        font-family: Georgia, serif;
    }
    .card p {
        font-size: 14px;
        font-family: Arial, sans-serif;
    }
    .card:hover {
        background: #f5f5f5;
        box-shadow: 5px 5px 12px #888;
    }
    .card button {
        background: #444;
        color: white;
        border: 1px solid #666;
        border-radius: 5px;
        padding: 6px 22px;
        margin-top: 8px;
        box-shadow: 2px 2px 5px #aaa;
    }
    .card button:hover {
        background: #666;
        color: white;
        box-shadow: 3px 3px 8px #888;
    }
</style>
</head>
<body>
    <header>
        <h1>My Styled Webpage</h1>
        <p>CSS3 Styling with HTML5 Canvas</p>
    </header>
    <nav>
        <a href="#">Home</a>
        <a href="#">About</a>
        <a href="#">Services</a>
        <a href="#">Contact</a>
    </nav>
    <hr>

```

```

<h2>Canvas Graphics</h2>
<canvas id="myCanvas" width="450" height="200"></canvas>
<script>
    var canvas = document.getElementById("myCanvas");
    var ctx = canvas.getContext("2d");
    ctx.fillStyle = "blue";
    ctx.fillRect(30, 30, 100, 60);
    ctx.fillStyle = "black";
    ctx.font = "12px Arial";
    ctx.fillText("Rectangle", 45, 110);
    ctx.beginPath();
    ctx.fillStyle = "red";
    ctx.arc(210, 60, 40, 0, 2 * Math.PI);
    ctx.fill();
    ctx.fillStyle = "black";
    ctx.fillText("Circle", 195, 120);
    ctx.beginPath();
    ctx.strokeStyle = "green";
    ctx.lineWidth = 4;
    ctx.moveTo(320, 30);
    ctx.lineTo(400, 90);
    ctx.stroke();
    ctx.fillStyle = "black";
    ctx.fillText("Line", 345, 120);
    ctx.fillStyle = "black";
    ctx.font = "bold 22px Arial";
    ctx.fillText("Nandu", 150, 170);
</script>
<hr>
<div class="container">
    <div class="card">
        <h3>Web Dev</h3>
        <p>HTML, CSS, JavaScript</p>
        <button>Learn</button>
    </div>
    <div class="card">
        <h3>Design</h3>
        <p>Graphics & UI/UX</p>
        <button>Learn</button>
    </div>
    <div class="card">
        <h3>Marketing</h3>
        <p>SEO & Social Media</p>
        <button>Learn</button>
    </div>
</div>
</body>
</html>

```